THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 3: **The World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal expansion of $\frac{3}{8}$ is: 1

(a) non-terminating non-repeating **(b)** non-terminating repeating
(c) terminating **(d)** cannot be determined

2. The decimal expansion of $\frac{11}{40}$ is: 1

(a) terminating **(b)** non-terminating non-repeating
(c) non-terminating repeating **(d)** cannot be determined

3. The number $0.\overline{45}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{11}{5}$ **(b)** $\frac{61}{110}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{6}{11}$

4. The number $0.35\overline{7}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{103}{225}$ **(b)** $\frac{161}{450}$ **(c)** $\frac{450}{161}$ **(d)** $\frac{9}{25}$

5. The number $1.01001000100001\ldots$ is: 1

(a) irrational **(b)** not a real number
(c) neither rational nor irrational **(d)** rational

6. A rational number in lowest terms has denominator $2^3 \times 5$. Its decimal expansion terminates after how many places? 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

7. The value of $(-8) \times (-7)$ is: 1

(a) -56 **(b)** 56 **(c)** 60 **(d)** 58

8. Which number is both rational and an integer? 1

(a) -7 **(b)** π **(c)** $\sqrt{2}$ **(d)** $\frac{3}{7}$

9. Who gave the first proof of the irrationality of $\sqrt{2}$ by contradiction (c. 400 BCE)? 1

(a) Lambert **(b)** Āryabhaṭa **(c)** Brahmagupta **(d)** Hippasus

10. A student claims: "$\sqrt{2} \times \sqrt{2}$ is irrational, as both numbers are irrational." The correct response is: 1

(a) False; $\sqrt{2} \times \sqrt{2} = 2$ is rational **(b)** True only for square roots
(c) True **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Express $0.\overline{6}$ in the form $\frac{p}{q}$. 2

12. Express $0.\overline{45}$ in the form $\frac{p}{q}$. 2

13. Express $0.1\overline{3}$ in the form $\frac{p}{q}$. 2

14. Express $0.\overline{18}$ in the form $\frac{p}{q}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{2}$ is irrational. 3

16. Prove that $\sqrt{3}$ is irrational. 3

17. Prove that $\sqrt{5}$ is an irrational number. 3

OR

Show that $3+\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational. 3

18. Express $2.3\overline{57}$ in the form $\frac{p}{q}$. 3

OR

Convert $0.1\overline{6}$ into the form $\frac{p}{q}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Prove that $\sqrt{2}$ is irrational. 5
(ii) Hence show that $3\sqrt{2}-5$ is irrational.

OR

(i) Express $0.\overline{27}$ and $1.2\overline{5}$ in the form $\frac{p}{q}$. 5
(ii) Write $0.\overline{9}$ as a fraction and comment.

20. **Case study:** The Ishango bone has a column of notches grouped as 11, 13, 17, 19. A merchant of Lothal gets 15 copper ingots for every 2 bags of spices. 5
(i) What do 11, 13, 17, 19 have in common? Name the next three numbers of this kind.
(ii) How many ingots will he get for 12 bags?
(iii) Are natural numbers closed under subtraction? Give an example.

OR

(i) Without long division, state which of $\frac{13}{80}$, $\frac{5}{12}$, $\frac{29}{343}$ have terminating decimals. 5
(ii) For the terminating one, find the decimal expansion.
(iii) Find the decimal of $\frac{5}{12}$ up to 5 places and state the repeating digit.

END OF QUESTION PAPER — **Total: 40 Marks**

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